CSE 4403

Assignment

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Section: 2

Topic: Ford Fulkerson Algorithm

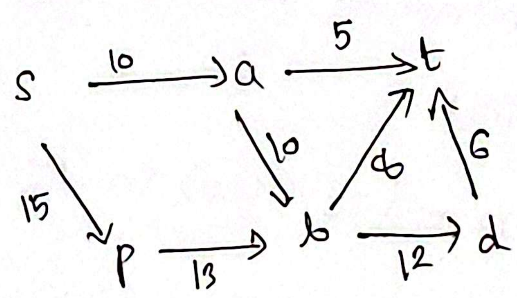
a) Description of Ford-Fulkerson Algorithm

Idea:

The Ford-Fulkerson algorithm finds the maximum flow in a flow network. It repeatedly finds path from source (s) to the sink (t) where additional flow can be pushed (called augmenting paths).

Each time, we increase the flow along this path by the minimum residual capacity on that path. This continues until no more augmenting paths exist.

Diagram



Each edge shows capacity. The algorithm will:

1. Find an s-t path (like $s \to a \to t$ or $s \to p \to b \to t$)

2. push flow = min (capacities along path)

3. Update residual graph (reduce capacity along used edges, add reverse edges).

Pseudocode:

```
FordFulkerson (G, s, t):

  For each edge (u,v) in G:
      flow(u,v) = 0

  while there exists an augmenting path p
  from s to t in residual graph:

      c_f(p) = minimum residual capacity of
      edges in p

      for each edge (u,v) in P:

          flow(u,v) = flow (u,v) + c_f(p)
          flow (v,u) = flow(v,u) - c_f(p)

  return sum of flows from s
```

Complexity:

Let

E = number of edges

V = number of vertices

C = maximum capacity among edges.

Then:

Time complexity: $O(E \times \text{max flow})$ in general. ~~If~~ Using Edmond Karp's algorithm

$O(VE^2)$

Space complexity: $O(V+E)$ (to store graph graph and residual network)

b) Proof of Correctness

- Each augmenting path or augmentation increases total flow.
- Flow never exceeds capacity (due to residual capacity checks)

- When no augmentation remains, by Max-Flow Min-cut theorem, the flow is maximum.

Hence algorithm terminates with maximum possible flow.

(c) Application Example

Problem Description:

Suppose you have multiple bus deports (sources) and destinations (sinks), connected by roads that can handle only a limited number of buses (capacity)

You want to find the maximum number of buses that can be scheduled ~~simultaneous~~ simultaneously without exceeding road limits.

This can be modeled as flow network:

- Each road has edge with capacity = number of buses that can pass.
- Depot = source node
- Destination = sink node

Then, Ford-Fulkerson gives the maximum feasible bus schedule

Implementation for this problem is attatched. Implementation is done using Edmond-Karp's algorithm